

GT For Compact Polyhedra

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1 Goldblatt-Thomason I

Definition 1. Let P, X be polyhedra, Σ a triangulation of X , and U an open subpolyhedron of P .

When Σ is equipped with the poset topology, we call $f : U \rightarrow \Sigma$ a **face up-reduction** $P \rightsquigarrow X$ if

- f is polyhedral, i.e. $f^{\leftarrow}(V)$ is an open subpolyhedron of U for any upward closed set $V \subseteq \Sigma$
- f is open

Theorem 2. A class of polyhedra C is modally definable if and only if it's closed under finite disjoint unions and face up-reductions, i.e.

(I) if $P, Q \in C$, then $P \sqcup Q \in C$

(II) if $P \in C$ and $P \rightsquigarrow X$, then $X \in C$.

Proof ($\implies$). Let C be a modally definable class, i.e. $C = \{P \in \mathbf{Pol}^K : P \models F\}$ for some set of formulas F .

Then if $P, Q \in C$, then $P \sqcup Q \models F$.

If $P \in C$ and $P \rightsquigarrow X$, then suppose $X \notin C$, i.e. there exists a formula $\varphi \in F$ such that $X \not\models \varphi$. In other words, there exists an evaluation $v : \text{Fmla} \rightarrow X$ and a point $x \in X$ such that $X, x \not\models \varphi$. We may select a triangulation S of X such that $v(\varphi)$ is a subtriangulation of S . Now call Σ the triangulation of X realizing the face up-reduction $P \rightsquigarrow X$.

From [?] we know that there exists a barycentric subdivision of Σ triangulating S .

Furthermore, we can extend any up-p-morphism $P \rightarrow \Sigma$ to an up-p-morphism $P \rightarrow \Sigma^{(k)}$ for any $k \in \mathbb{N}$.

This means that we have an up-p-morphism $P \rightarrow \Sigma^{(k)} \rightarrow S$, therefore $P \not\models \varphi$, leading to a contradiction. $\square$

Proof ($\impliedby$). Let C be a class of polyhedra closed under (I) and (II), call $\mathcal{L} := \text{Log}(C)$ and call $C' := \{P \in \mathbf{Pol}^K : P \models \mathcal{L}\}$

Clearly $C \subseteq C'$. Let $X \in C'$, we want to show that $X \in C$.

We know that $X \models \mathcal{L}$, and let $\Sigma \in \text{Tr}(P')$.

Call v_i the vertices of Σ . We know that $\bigsqcup_i (\uparrow v_i) \rightarrow \Sigma$ is a p-morphism.

Consider $\chi_i := \chi(\uparrow v_i)$ the Jankov-Fine formula for the (rooted) subposet.

As $(\uparrow v_i) \not\models \chi_i$, we know that $\Sigma \not\models \chi_i$, as all $(\uparrow v_i)$ are generated subframes of Σ .

In particular $X \rightarrow \Sigma$, thus $X \not\models \chi_i$, meaning none of the χ_i are elements of $\mathcal{L}$.

This means that for each χ_i there exists some $P_i \in C$ such that $P_i \not\models \chi_i$, thus $P_i \rightarrow (\uparrow v_i)$.

This means that $P := \bigsqcup_i P_i$ admits an up-p-morphism to $\bigsqcup_i (\uparrow v_i)$, and we can take the composition with $\bigsqcup_i (\uparrow v_i) \rightarrow \Sigma$ to find $P \rightsquigarrow X$. By (I) we know $P \in C$, and by (II) $X \in C$. $\square$

2 Goldblatt-Thomason II

Now we want to show that (II) is equivalent to

(II') $P \in \mathcal{C}$ and there exists an open polyhedral map $P \supseteq U \rightarrow X$ for U an open subpolyhedron of P , then $X \in \mathcal{C}$.

It's clear that (II') implies (II) as X admits an open polyhedral map $X \twoheadrightarrow \Sigma$ to any of its triangulations. It is less clear why (II) implies (II')

We will split this proof into two separate parts.

Lemma 3. *Let $f : K \rightarrow K'$ be a map of simplicial complexes. Then f is open if and only if $|f|$ is open.*

Proof. Recall that if $|\Sigma|$ is the realization of Σ then there exists an open polyhedral map $\pi : |\Sigma| \rightarrow \Sigma$, allowing us to draw the diagram

$$\begin{array}{ccc} |\Sigma'| & \dashrightarrow & |\Sigma| \\ \pi' \downarrow & & \downarrow \pi \\ \Sigma' & \longrightarrow & \Sigma \end{array}$$

One side of the implication is simple: If $|f|$ is open, then suppose $U' \subseteq \Sigma'$ is an open subcomplex, i.e. U' is upward-closed. $V' = \pi'^{\leftarrow}(U')$ is an open subpolyhedron of $|\Sigma'|$.

$f(U) = \pi|f|(V)$ is open.

Conversely, suppose $V' \in |\Sigma'|$ is open. Then by [?] there exists an iterated baricentric subdivision $\Sigma'^{(k)}$ of Σ' extending Σ' and covering V' .

We may extend f to $\Sigma'^{(k)}$, and by the previous lemma it will be open. We may do the same to $|f|(V)$, thus $|f|(V) = \pi^{(l)\leftarrow} f^{(k)}(\pi^{(k)}(V))$ which is open. $\square$

2.1 Nerving

This is the correct moment to fix some notations about nerves as the following lemmas will use it extensively.

Definition 4. *Let P be a compact polyhedron, $\Sigma \in \text{Tr}(P)$. Then $N(\Sigma)$ is the poset consisting of chains $(\sigma_0 < \dots < \sigma_k)$ for $\sigma_i \in \Sigma$ ordered by inclusion. This constitutes the face poset of a simplicial complex triangulating P . We call $\bar{\sigma}$ elements of $N(\Sigma)$ and in particular if $\sigma_i \in \Sigma$, then $\bar{\sigma}_i = (\sigma_i)$ (the chain with one element containing it).*

When taking iterated nerves we say $\bar{\sigma}^1 = \bar{\sigma}$ and $\bar{\sigma}^{n+1} = \overline{\bar{\sigma}^n}$

Lemma 5. *If $S \subseteq \Sigma$ is a closed subcomplex of Σ , then $N(S) \subseteq N(\Sigma)$ is a subcomplex of $N(\Sigma)$. Furthermore S and $N(S)$ triangulate the same closed subpolyhedron of P*

Proof. $S \subseteq \Sigma$ is a subcomplex if and only if it's downward closed. Let $\sigma \in \Sigma$, then $\bar{\sigma} = (\sigma) \in N(S)$ If $\bar{\tau} \in N(S)$, then $\bar{\tau} = (\tau_0 < \dots < \tau_k)$ for $\tau_0, \dots, \tau_k \in S$. This means that any $(\tau_i < \dots < \tau_j) \in \downarrow \bar{\tau}$ is also comprised of elements of S , thus a member of $N(S)$.

Moreover recall that $|N(S)| = |S|$ if S is a simplicial complex. We call $N(S) := \bar{S}$ following the notation described above. $\square$

Then for all $x \in X$, $\overline{\pi(x)} \leq \pi_1(x)$ in $N(\Sigma)$.

We know that there exists a polyhedral open map $\pi : |\Sigma| \rightarrow \Sigma$, then consider $\pi^{\leftarrow}(U) := |U| \subseteq |\Sigma|$. Call $\pi_1 : |\Sigma| \rightarrow N(\Sigma)$ the open polyhedral map onto the nerve. Then $\sigma \in \pi_1(|U|)$ if and only if $\sigma \in \uparrow \bar{\sigma}_i$ for some $\sigma_i \in U$.

Conversely, if $\sigma \in \uparrow \overline{\sigma_i}$ for some $\sigma_i \in U$, then $\sigma = (\sigma_i < s_1 < \dots < s_k)$. As U is upward-closed, each s_i is an element of U as well. This means each $\overline{s_i}$ is a vertex of σ . $\pi_1^{\leftarrow}(\overline{s_i})$ is the center of a simplex in U , thus in particular $\pi_1^{\leftarrow}(\sigma) \in |U|$.

Let $B = \downarrow \uparrow \overline{\Sigma_C}^2 \subseteq N^2 \Sigma$
 Then $|B| \subseteq U$.

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We want to show that $\tilde{f}$ is open and surjective:

It is open since it's the restriction of an open map to an open subset.

Note that

$$\tilde{f}(U) = \bigcup_i \uparrow_{N^2} \downarrow \uparrow_N N(f)(\bar{p}_i) = \bigcup_i \uparrow_{N^2} \downarrow \uparrow_N \bar{v}_i$$

We can show that $\bigcup_i \downarrow \uparrow_N \bar{v}_i$ covers $N(\Sigma)$: Let $\bar{\sigma}$ be a simplex of $N(\Sigma)$.

Then $\bar{\sigma} = (\sigma_0 < \dots < \sigma_k)$. If $\sigma_0 = v_i$ is a vertex, then $\bar{\sigma} \in \uparrow v_i \subseteq \downarrow \uparrow v_i$. If σ_0 is not a vertex, then consider any $w \in V_{\sigma_0}$: $w < \sigma_0$, thus

$(\sigma_0 < \dots < \sigma_k) \leq (w < \sigma_0 < \dots < \sigma_k)$ thus $\bar{\sigma} \in \downarrow(\uparrow w)$.

As $\downarrow \uparrow v_i$ cover Σ , $\uparrow \downarrow \uparrow v_i \supseteq \downarrow \uparrow v_i$ cover Σ as well, and form an open cover, thus U is open, thus $\tilde{f}$ is open.

by the previous lemma F is open since $\tilde{f}$ is. □

Proposition 11. *Let P, X be compact polyhedra, Σ a triangulation of X and $f : P \rightarrow \Sigma$ an up- p -morphism defined on an open subpolyhedron $U \subseteq P$.*

Then there exists a triangulation T of P and a closed subcomplex T_U such that

- $|T_U| \subseteq U$
- $f|_{T_U} : T_U \rightarrow \Sigma$ is a surjective p -morphism

Proof. Following the steps of the previous proof, choosing $T_U := \downarrow \uparrow_{N^2} \downarrow \uparrow_N p_i$ in an appropriate triangulation is enough. To ensure that $|T_U| \subseteq U$ we need to choose a triangulation T such that p_i are vertices and it triangulates f . Then pick $\tilde{T} = N^2 T$

By proposition 9, p_i is a closed subcomplex in T_U , thus $\uparrow \downarrow p_i$ is a closed subcomplex contained in $N^2(T_U)$, thus $\uparrow \downarrow(\uparrow \downarrow p_i)$ is a closed subcomplex contained in $N^2(N^2(T_U)) = N^4(T_U) \subseteq N^2 \tilde{T}$,

then we apply the previous proposition to it. □